

ISA ALTUNOGLU

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Education

Karabuk University

Oct. 2023 – Present

Bachelor of Science in Computer Engineering (100% English Curriculum) – GPA: 3.27/4.00

Karabuk, Turkey

Experience

Project Manager & Technical Member

Present

Hicaz Hyperloop Team (Teknofest)

Karabuk, Turkey

- Managed project milestones and documentation while engineering **Communication protocols** and **GUI** systems for vehicle telemetry.
- Developed innovative software solutions for real-time data visualization and system control.

Volunteer Backend Developer

Jan. 2025 – Jun. 2025

Karabuk Iron and Steel Institute (KARDEMİR)

Karabuk, Turkey

- Architected and developed a backend automation system using **Go (Golang)** to track worker operations.
- Designed high-performance REST APIs and utilized Postman for rigorous API testing and documentation.
- Collaborated with a cross-functional team to integrate backend services with frontend interfaces.

Coordination Unit Member

Present

GEPTEK Student Club

Karabuk, Turkey

- Orchestrated digital information flow and managed official announcements for club events.
- Facilitated internal coordination to ensure synchronized team operations.

Projects

Frost Risk Prediction (TÜBİTAK 2209-A) | Project Lead, ESP32, LSTM

Present

- Spearheaded a research project to predict frost risks, architecting a hybrid **IoT (ESP32)** and **Deep Learning (LSTM)** system.
- Designed the embedded firmware to calculate "Wet Bulb Temperature" and forecast micro-climate data.
- Leading the entire solution architecture to predict agricultural frost risks 3 hours in advance.

Hydroponic Agriculture System (TÜBİTAK 2209-A) | Software & IoT Developer

Present

- Engineered the **Software and IoT architecture** integrating **Raspberry Pi** and **Arduino** for automated nutrient control.
- Implemented real-time sensor data acquisition pipelines to support AI-driven disease detection.
- Integrated Deep Learning models for plant disease detection into the embedded control loop.

Drone Control Station Software | Qt Creator, QGroundControl, C++

Jun. 2024 – Aug. 2024

- Customized QGroundControl open-source software using Qt Creator for a specialized rescue drone.
- Developed custom plugins in C++ to adapt the interface for earthquake survivor detection missions.

Technical Skills

Languages: Python, Go (Golang), C, C++, Java, Dart

AI & Data Science: PyTorch, TensorFlow, Keras, LSTM, EfficientNet, OpenCV, LLM Integration (OpenAI/Gemini)

Web & Cloud: Next.js, React, Firebase, REST APIs

Embedded & Tools: ESP32, Raspberry Pi, Arduino, Git, GitHub, Postman, Qt Creator, Linux, QGroundControl

Design & Manufacturing Blender 3D Printing

Spoken Languages: Turkish (Native), English (B2)

Professional Development

Leadership & Management

Oct 2025 – Present

Demonstrated ability to lead technical teams and manage research projects

Karabuk University

- Leading Hicaz Hyperloop Team for Teknofest competition as Project Manager with multidisciplinary engineering team.
- Managing TÜBİTAK 2209-A research project as Project Lead for frost risk prediction system.
- Coordinating GEPTEK Student Club activities and managing digital communication channels.